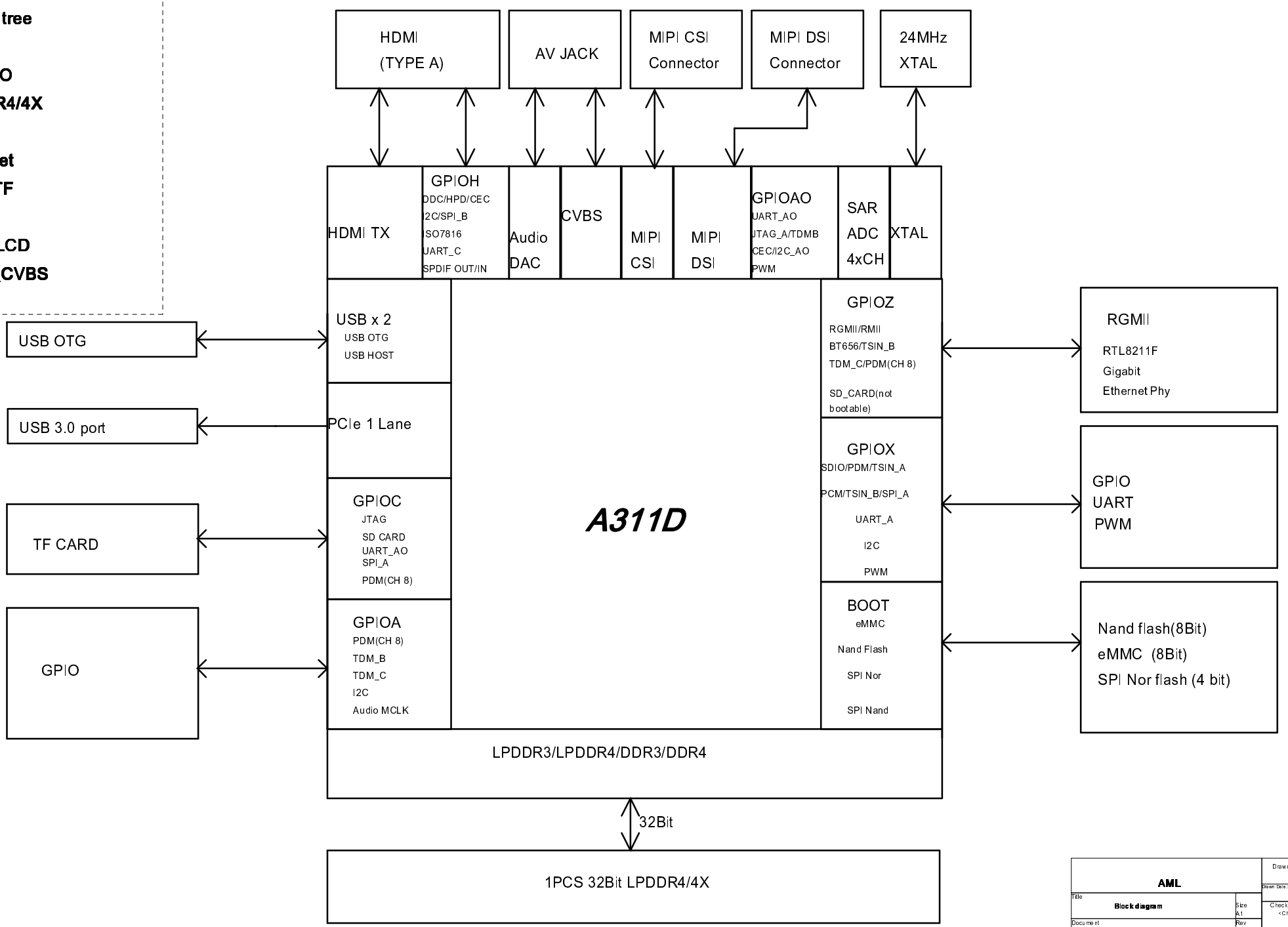
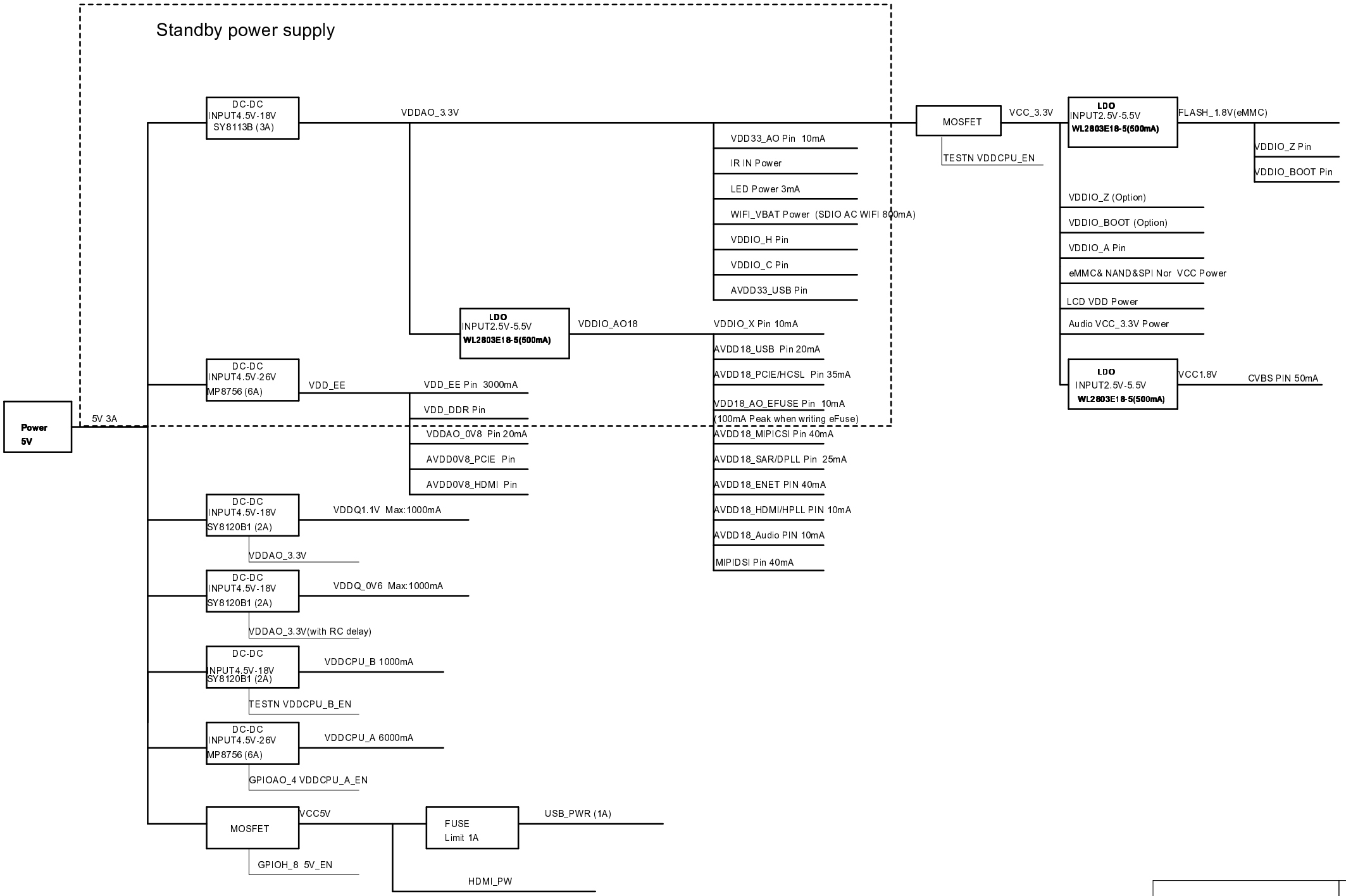
A311D Diagram

PAGE INDEX

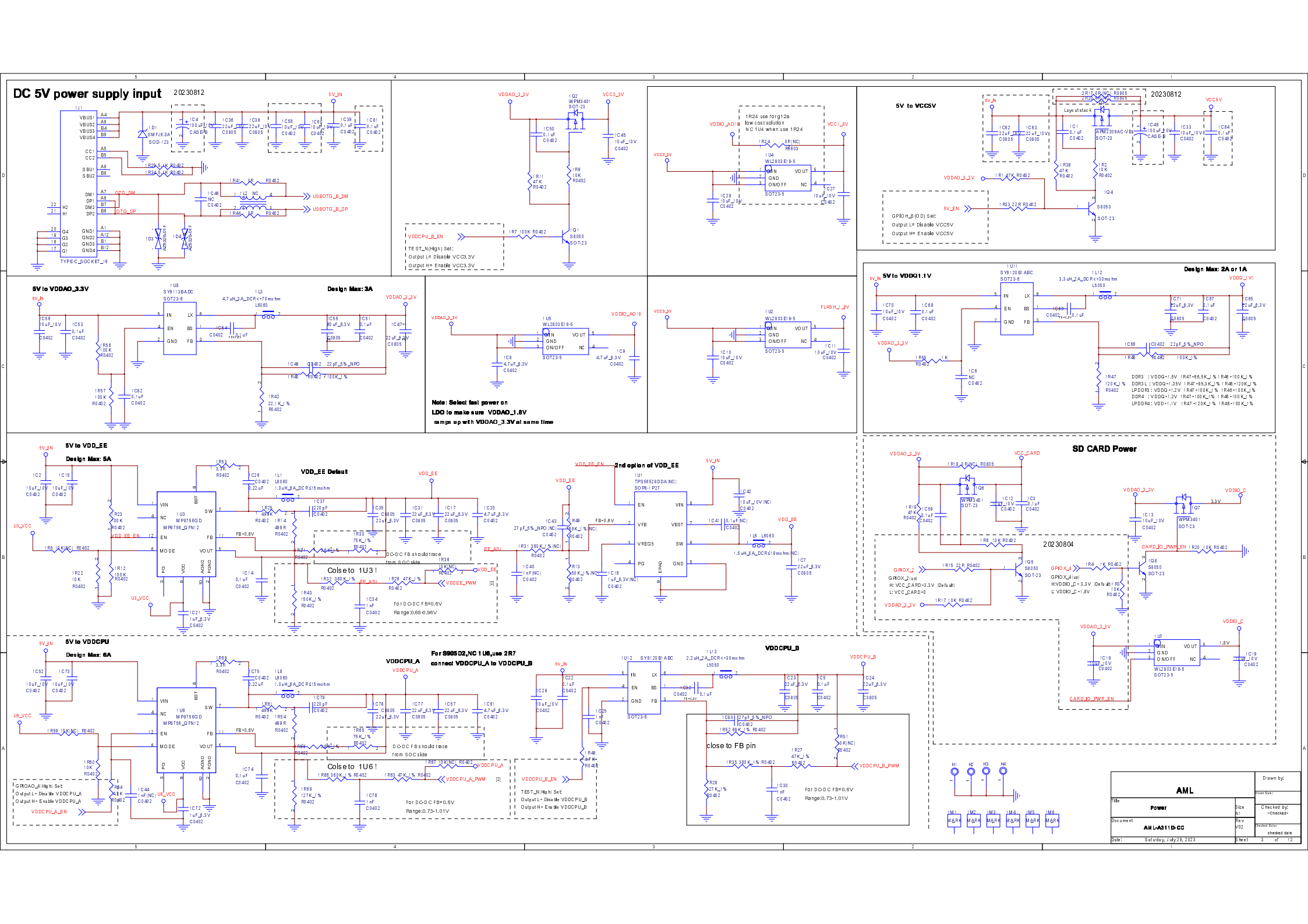
00- Block diagram
00- Power tree
01- Power
02- CPU I/O
03- LPDDR4/4X
04- Flash
05- Ethernet
06- USB_TF
07- GPIO
08- CAM_LCD
09- HDMI_CVBS
10- Audio



AML		Drawn by:
Title: Block diagram	Size: A1	Drawn Date:
Document: AML-A311D-CC	Rev: V02	Checked by: <Checked>
Date: Saturday, July 29, 2023	Sheet 1 of 12	Checked Date: checked date

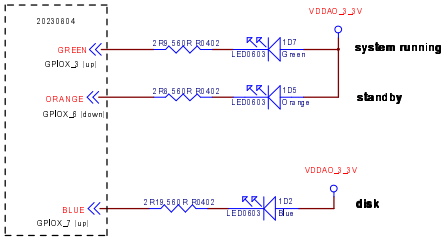


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Document: AML-A311D-CC	Rev: V02	Checked by: <Checked>
Date: Saturday, July 29, 2023	Sheet: 2 of 12	Checked Date: checked date

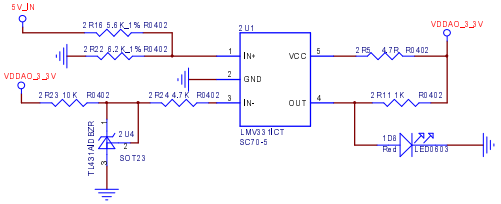

DC 5V power supply input
20230812
5V_IN
TYPE-C_SOCKET_16
USBOTG_B_DM
USBOTG_B_DP
OTG_DM
OTG_DP
5V to VDDAO_3.3V
Design Max: 3A
SY8113BADC
SOT23-6
VDDAO_3_3V
Note: Select fast power on
LDO to make sure VDDAO_1.8V
ramps up with VDDAO_3.3V at same time
WL2803E18-5
SOT23-5
VDDAO_1V8
FLASH_1_8V
5V to VDD_EE
Design Max: 5A
VDD_EE Default
2nd option of VDD_EE
MP8756GD
VDD_EE
VDD_EE_PWM
Colse to 1U3 !
DC-DC FB should trace from SOC side
For DC-DC FB=0.6V
Range:0.68~0.96V
5V to VDDCPU
Design Max: 6A
VDDCPU_A
VDDCPU_B
For S905D2_NC 1 U6,use 2R7
connect VDDCPU_A to VDDCPU_B
Colse to 1U6 !
close to FB pin
VDDCPU_A_PWM
VDDCPU_B_PWM
For DC-DC FB=0.6V
Range:0.73~1.01V
5V to VCC5V
APM2309AC-VB
SOT-23
VCC5V
GPIO H_B(OD) Set:
Output L= Disable VCC5V
Output H= Enable VCC5V
5V to VDDQ1.1V
SY8120BIABC
Design Max: 2A or 1A
DDR3 : VDDQ=1.5V
DDR3L : VDDQ=1.35V
LPDDR3 : VDDQ=1.2V
DDR4 : VDDQ=1.2V
LPDDR4 : VDD=1.1V
SD CARD Power
20230804
CARD_IO_PWR_EN
GPIOX_2 Set:
H: VCC_CARD=3.3V (Default)
L: VCC_CARD=0
GPIOX_4 Set:
H:VDDIO_C=3.3V (Default)
L: VDDIO_C=1.8V
AML
Power
AML-A311D-CC
Saturday, July 29, 2023
Sheet 3 of 12

STATUS LED

Power attach: Orange on, Green off, Blue don't care

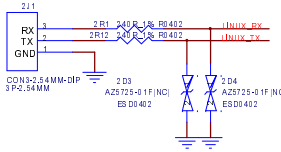


POWER LED



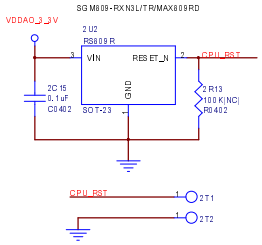
Red LED light off when 5V_IN<4.75V

For linux debug

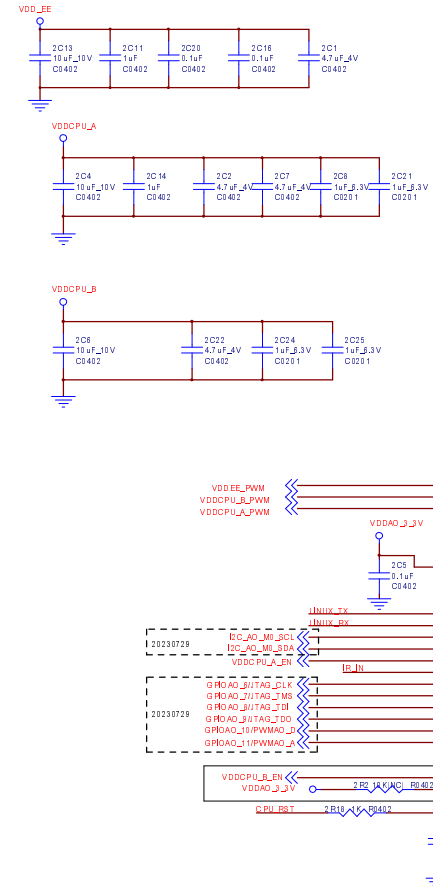


mark signal name on PCB

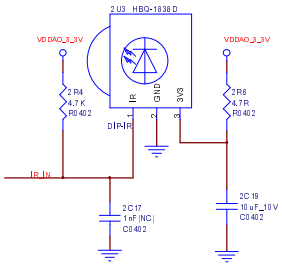
CPU reset



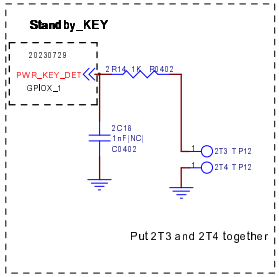
Put 2T1 and 2T2 together



Remote control receiver



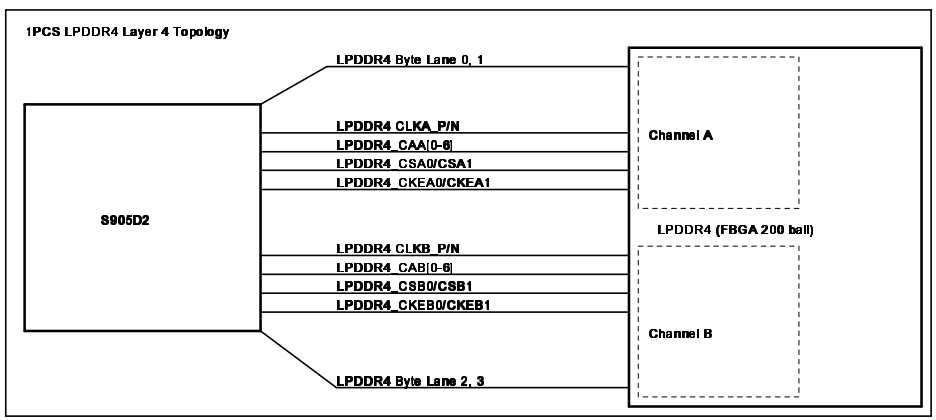
Stand by_KEY



Put 2T3 and 2T4 together

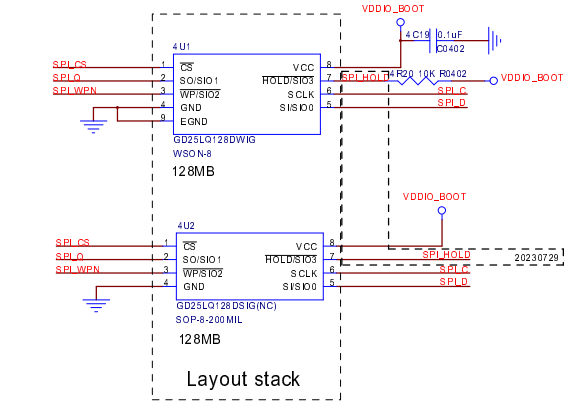
Heat sink

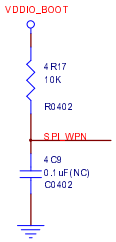
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Title: CPU I/O		Size: A1	Checked by: <Checked>
Document: AML-A311D-CC		Rev: V02	Checked Date: checked date
Date: Saturday, July 29, 2023	Sheet 4 of 12		

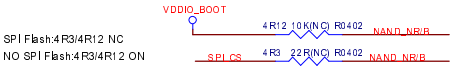


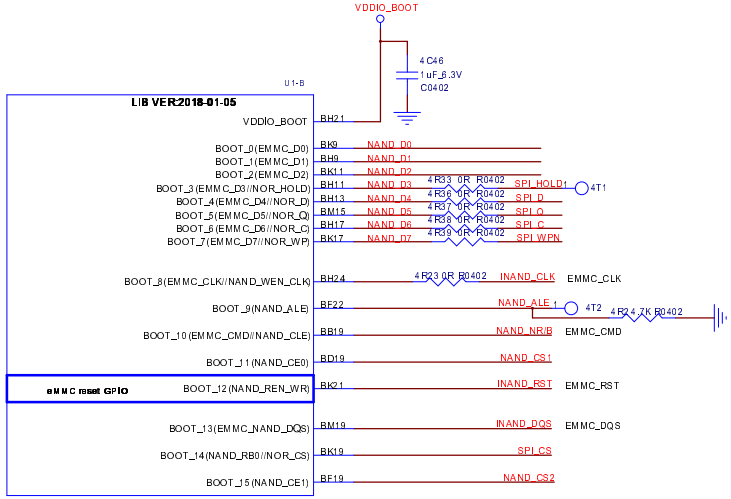
			Drawn by:
	AML		Shared Data:
Title	LPDDR4/4X	Size A1	Checked by: <Checked>
Document	AML-A311D-CC	Rev V02	Checked Date: checked date
Date:	Saturday, July 29, 2023	Sheet	5 of 12

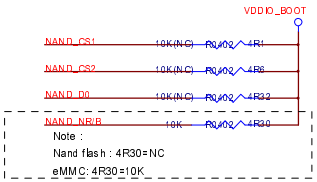
SPI NOR FLASH

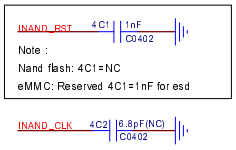


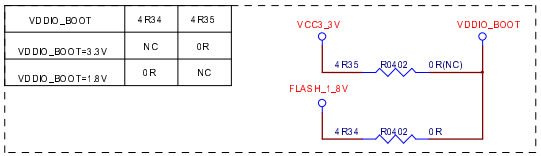


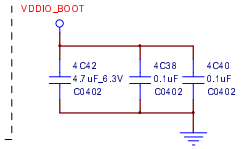


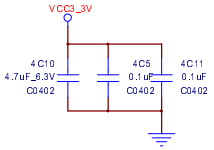












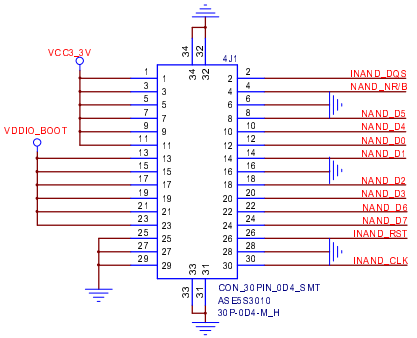
Close to 4J1 VDDIO pin

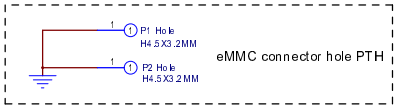
Reserved for update.



Put 4T10 and 4T11 together

External eMMC module



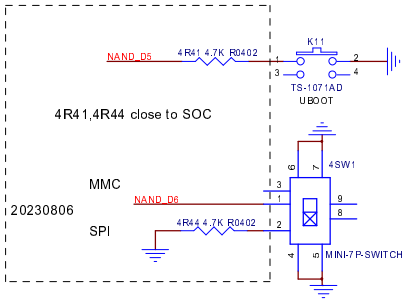


Power on config

BOOT_4[NAND_D4]: 0= SPI NAND Boot first

BOOT_5[NAND_D5]: 0= USB Boot first

BOOT_6[NAND_D6]: 0= SPI NOR Boot first



Title	Flash	Size A1
Document	AML-A311D-CC	Rev V02
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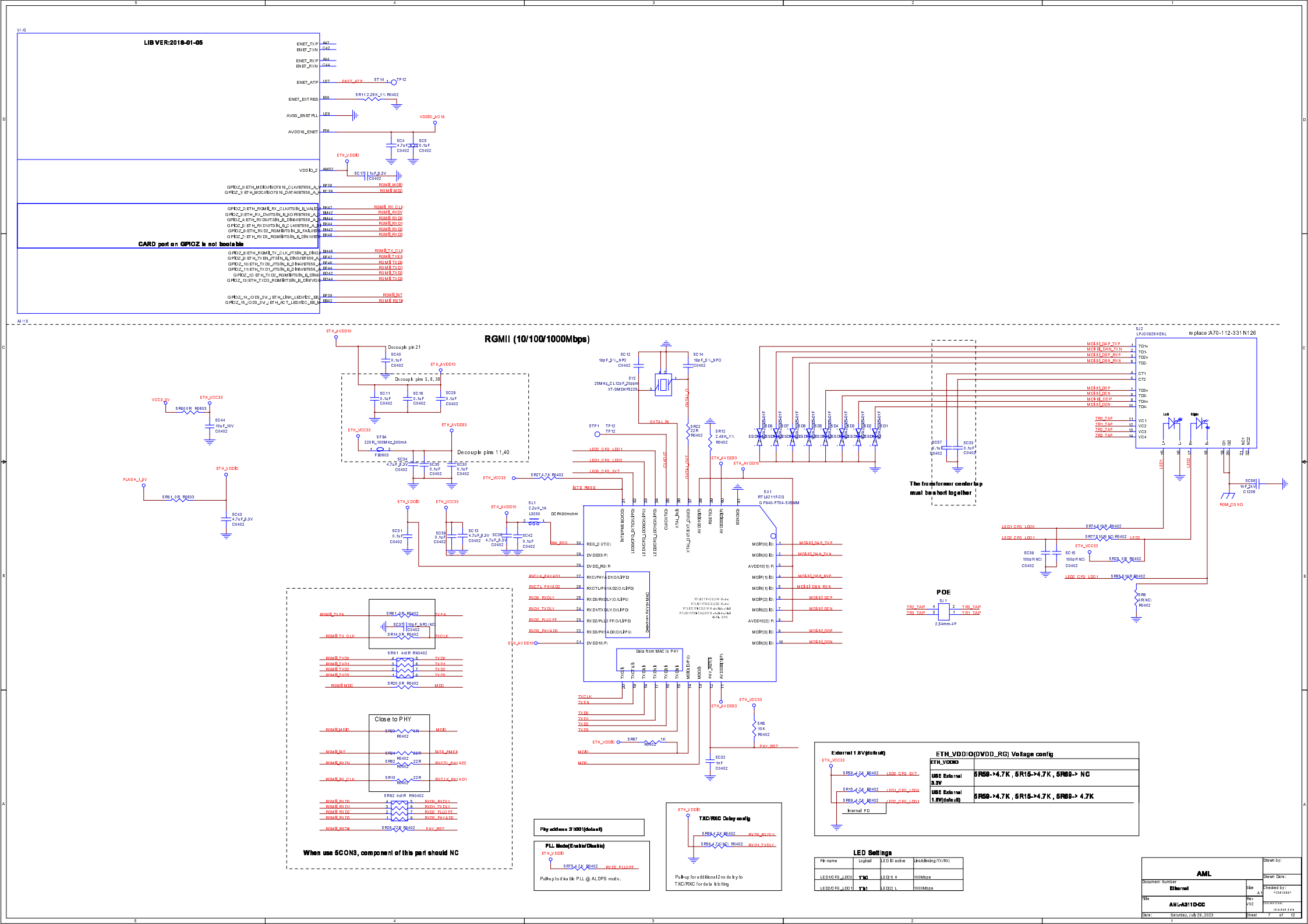
AML

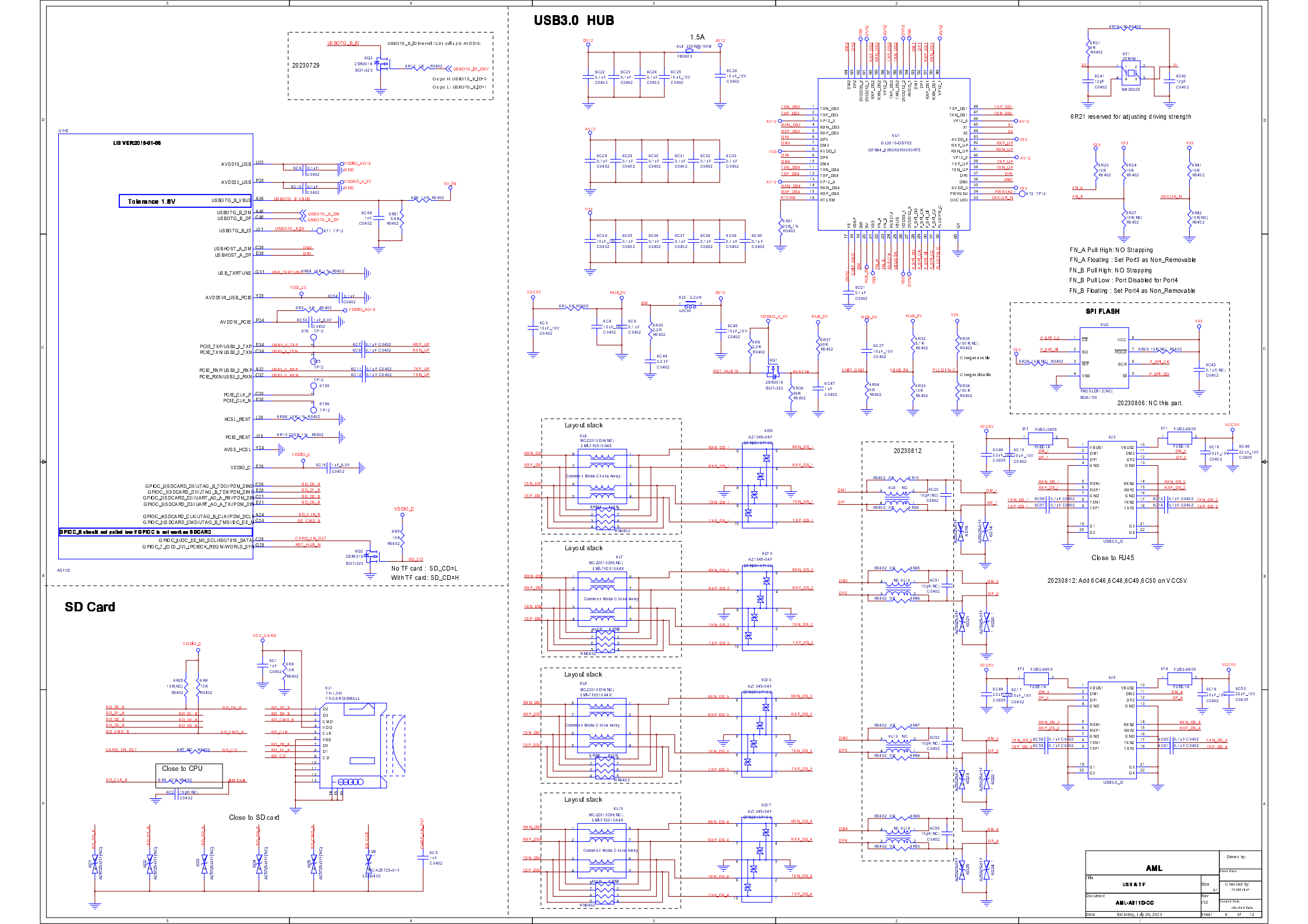
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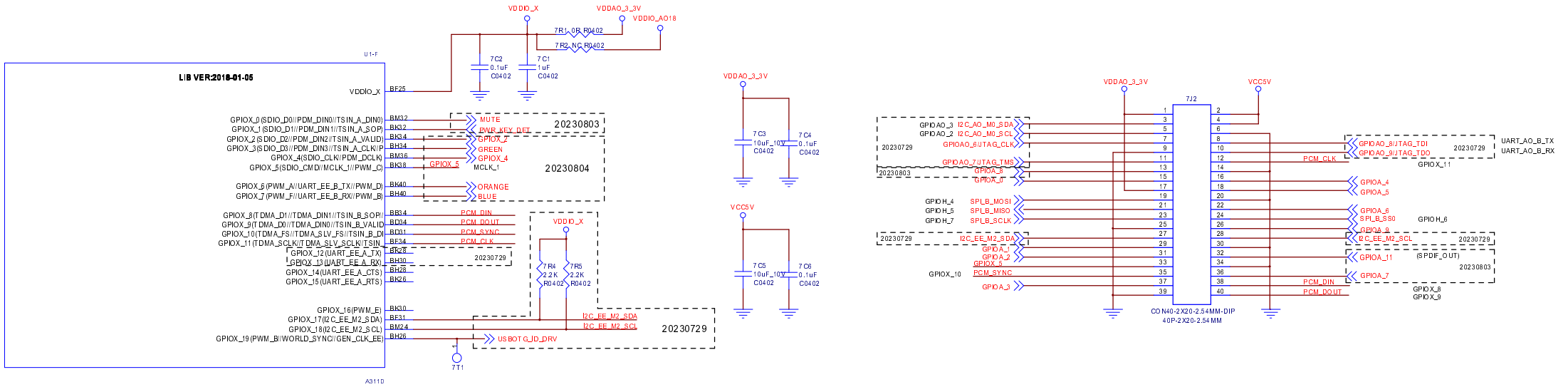
Drawn Date:

Checked by: <Checked>

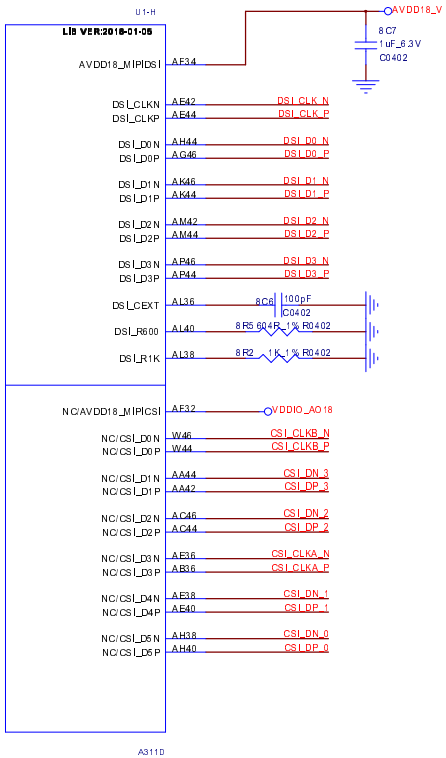
Checked Date: checked date


LIB VER:2018-01-05
CARD port on GPIOZ is not bootable
RGMII (10/100/1000Mbps)
Decouple pin 21
Decouple pins 3, 8, 38
Decouple pins 11,40
The transformer center tap must be short together
POE
When use 5CON3, component of this part should NC
Close to PHY
Data from MAC to PHY
Play address 3'b001(default)
PLL Mode(Enable/Disable)
Pull-up to disable PLL @ ALDPS mode.
TXC/RXC Delay config
Pull-up for additional 2ns delay to TXC/RXC for data latching
External 1.8V(default)
ETH_VDDIO(DVDD_RG) Voltage config
ETH_VDDIO
USE External 3.3V
5R59->4.7K , 5R15->4.7K , 5R69-> NC
USE External 1.8V(default)
5R59->4.7K , 5R15->4.7K , 5R69-> 4.7K
LED Settings
Pin name
Logical
LED Is active
Link/Blinking:TX/RX
LED1/CFG_LED0
LED2/CFG_LED1
100Mbps
1000Mbps
AML
Ethernet
AML-A311D-CC
Drawn by:
Drawn Date:
Checked by:
Size
A1
Rev
V02
Date:
Saturday, July 29, 2023
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7
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12

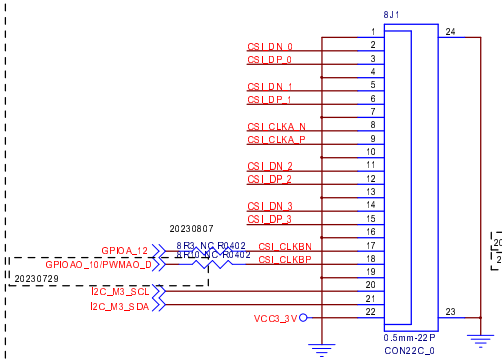

USB3.0 HUB
SD Card
LIB VER:2018-01-05
Tolerance 1.8V
GPIOC_8 should not pulled low if GPIOC is not work as SDCARD
No TF card : SD_CD=L
With TF card: SD_CD=H
Close to CPU
Close to SD card
1.5A
SPI FLASH
20230729
20230812
20230806: NC this part.
20230812: Add 6C46,6C48,6C49,6C50 on VCC5V
Close to RJ45
6R21 reserved for adjusting driving strength
FN_A Pull High: NO Strapping
FN_A Floating : Set Port3 as Non_Removable
FN_B Pull High: NO Strapping
FN_B Pull Low : Port Disabled for Port4
FN_B Floating : Set Port4 as Non_Removable
Layout stack
AML
USB&TF
AML-A311D-CC
Saturday, July 29, 2023

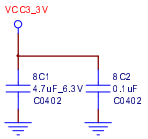


AML			Drawn by:
Title: GPIO		Size: A1	Drawn Date:
Document: AML-A311D-CC		Rev: V02	Checked by: <Checked>
Date: Saturday, July 29, 2023	Sheet 9 of 12		Checked Date: checked date

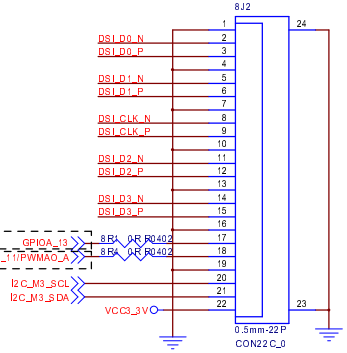


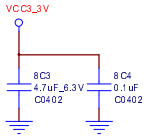
CAMERA



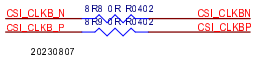


DISPLAY





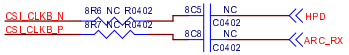
For Multi Cameras



20230807
A311D: 8R3,8R10 -- NC; 8R8,8R9 -- 0 R
S905D3: 8R3,8R10 -- 0R; 8R8,8R9 -- NC

20230804

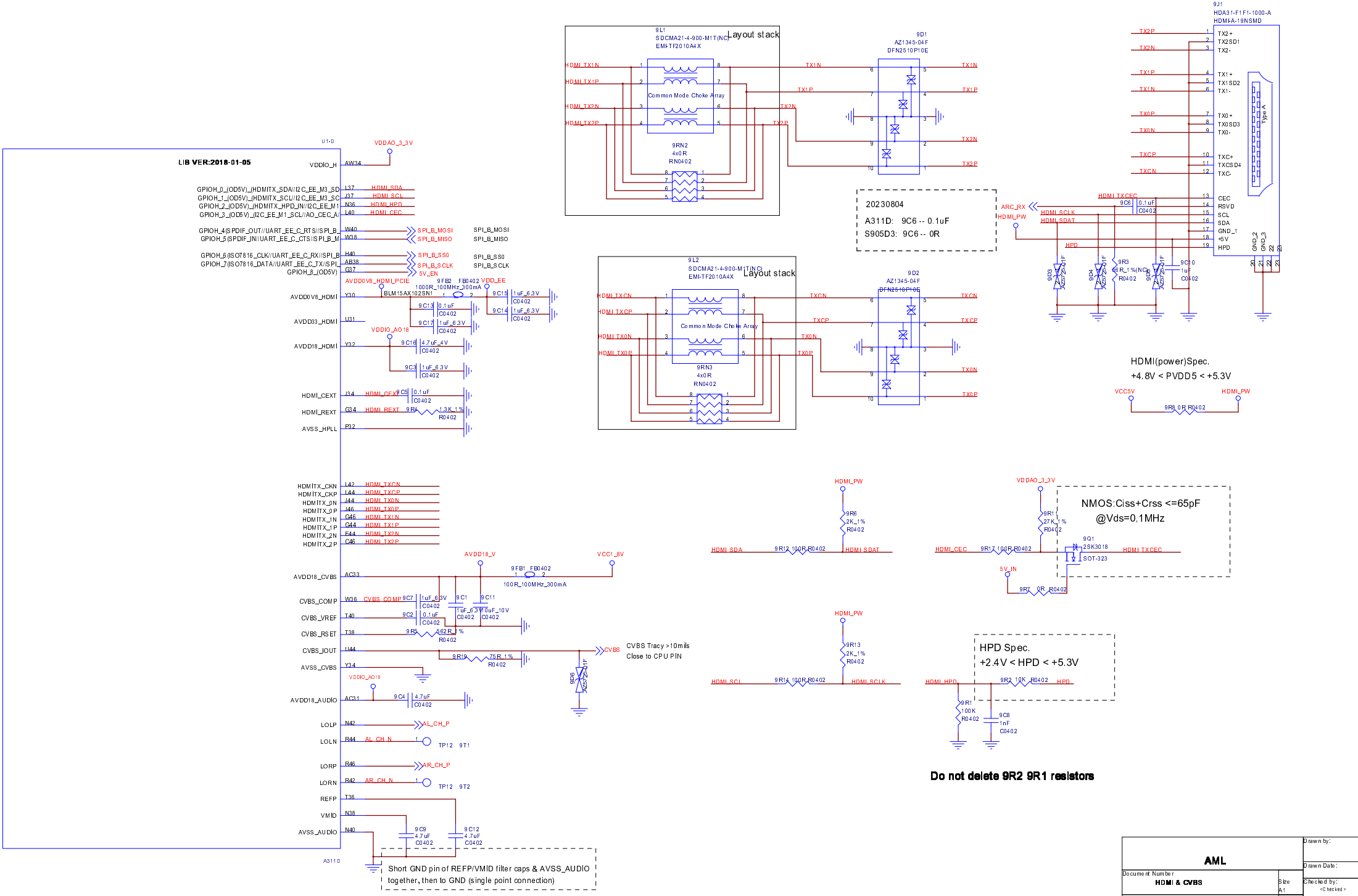
For S905D3 HDMI HPD/ARC

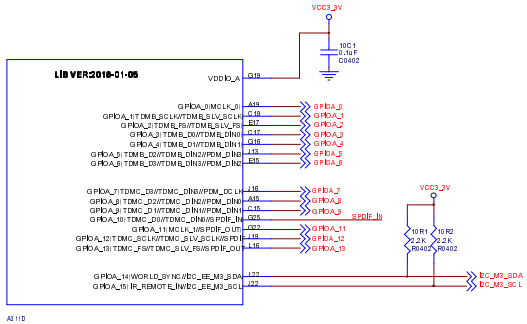


A311D: 8R6/8R7 -- NC
8C5/8C8 -- NC
S905D3: 8R6/8R7 -- 0R
8C5/8C8 -- 1uF

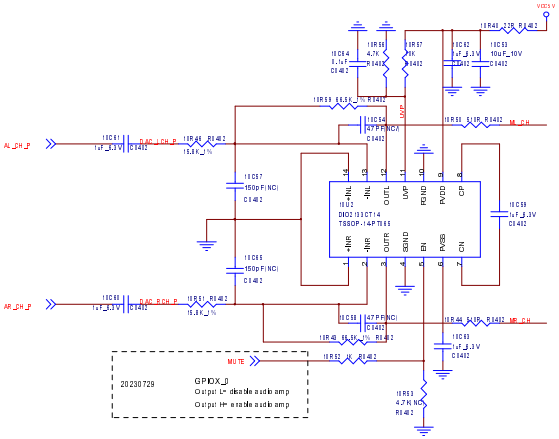
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Title: CAM_LCD	Size: A1	Drawn Date:
Document: AML-A311D-CC	Rev: V02	Checked by: <Checked>
Date: Saturday, July 29, 2023	Sheet 10 of 12	Checked Date: checked date

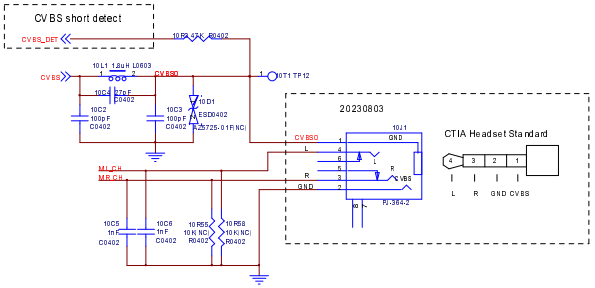
HDMI output

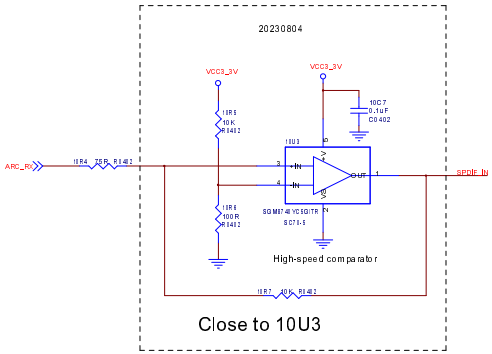

LIB VER:2018-01-05
Layout stack
Common Mode Choke Array
20230804
A311D: 9C6 -- 0.1uF
S905D3: 9C6 -- 0R
HDMI(power)Spec.
+4.8V < PVDD5 < +5.3V
NMOS:Ciss+Crss <=65pF
@Vds=0,1MHz
HPD Spec.
+2.4V < HPD < +5.3V
Do not delete 9R2 9R1 resistors
CVBS Tracy >10mils
Close to CPU PIN
Short GND pin of REFP/VMID filter caps & AVSS_AUDIO together, then to GND (single point connection)
AML
HDMI & CVBS
AML-A311D-CC
Saturday, July 29, 2023



CVBS out







AML			Drawn by:
Title: Audio		Size: A1	Checked by:
Document: AML-A311D-CC		Rev: V02	Checked date:
Date: Saturday, July 29, 2023		Sheet: 12 of 12	